

o. Prerequisites

- o.1. Differential and integral calculus of one and several variables
- o.2. Linear algebra: vector spaces, linear maps, inner products, dual space
- o.3. Measure theory: null sets, equality “almost everywhere”, integrable functions
- o.4. General topology: compactness, continuity, density
- o.5. *All norms in finite dimension are equivalent*

1. Basic integration results

1.1. Definitions.

The integral of positive functions is always defined and belongs to $\mathbf{R}^+ \cup \{+\infty\}$. A function $f : \Omega \rightarrow \mathbf{R}$ is *integrable* if $\int_{\Omega} |f| < \infty$. The set of all integrable functions on Ω is denoted $L^1(\Omega)$. Two functions of $L^1(\Omega)$ that coincide almost everywhere are considered as the same function. Using the norm $\|f\|_{L^1(\Omega)} := \int_{\Omega} |f|$, this set is a normed vector space (and it turns out to be complete).

1.2. Theorem (monotone convergence, Beppo Levi)

Let f_n be an increasing sequence of positive L^1 functions such that $\sup_n \int f_n < \infty$. Then $f_n(x)$ converges almost everywhere to a finite limit $f(x)$ such that $f \in L^1(\Omega)$, $\|f_n - f\|_{L^1} \rightarrow 0$ and $\int f_n \rightarrow \int f$.

1.3. Theorem (dominated convergence, Lebesgue)

Let f_n be a sequence of L^1 functions converging pointwise a.e. to a function $f(x)$. Assume there exist a function $h \in L^1(\Omega)$ such that $|f_n(x)| \leq h(x)$ a.e. Then $f \in L^1(\Omega)$, $\|f_n - f\|_{L^1} \rightarrow 0$ and $\int f_n \rightarrow \int f$.

1.4. Theorem (density)

The set $C_c(\Omega)$ of compactly supported continuous functions is dense in $L^1(\Omega)$.

1.5. Theorem (Tonelli)

If $f : \Omega_1 \times \Omega_2 \rightarrow \mathbf{R}^+ \cup \{+\infty\}$ then

$$\iint_{\Omega_1 \times \Omega_2} f(x, y) dx dy = \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) dy \right) dx = \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) dx \right) dy.$$

in particular, if one of these three integrals is $+\infty$, then the other two also are also. (In words: the sum of positive numbers, be it finite or infinite, does not depend on the order in which they are summed.)

1.6. Theorem (Fubini)

If $f \in L^1(\Omega_1 \times \Omega_2)$ then the three members in the formula above are well-defined and they are equal. In particular, the fact that the second member is well defined implies that

- The function $y \mapsto \int_{\Omega_1} f(x, y) dx$ belongs to $L^1(\Omega_2)$ for a.e. x
- The function $x \mapsto \int_{\Omega_2} f(x, y) dy$ thus defined belongs to $L^1(\Omega_1)$.

1.7. Theorem (change of variables)

Let $\varphi : \Omega_1 \rightarrow \Omega_2$ a diffeomorphism between two domains of $\mathbf{R}^n$ and J_{φ} its jacobian.

- If $f : \Omega_2 \rightarrow \mathbf{R}^+ \cup \{+\infty\}$ then

$$\int_{\Omega_2} f(y) dy = \int_{\Omega_1} f(\varphi(x)) |J_{\varphi}(x)| dx$$

where both sides have a meaning in $\mathbf{R}^+ \cup \{+\infty\}$.

- If $f : \Omega_2 \rightarrow \mathbf{R}$ then $f(x) \in L^1(\Omega_2)$ if and only if $f(\varphi(x)) J_{\varphi}(x) \in L^1(\Omega_1)$, and in that case both members of the previous equation are equal.

2. L^p spaces

2.1. **Definition.** Let us define the following numbers (which may be infinite)

$$\|f\|_{L^p(\Omega)} := \left(\int_{\Omega} |f|^p \right)^{1/p} \quad \text{for } 1 \leq p < \infty$$

$$\|f\|_{L^\infty(\Omega)} := \inf \left\{ M ; |f(x)| \leq M \text{ a.e. on } \Omega \right\}$$

The space $L^p(\Omega)$ for $1 \leq p \leq \infty$ is the set of functions such that $\|f\|_{L^p(\Omega)} < \infty$. To each $p \in [1, \infty]$ we associate its *conjugate exponent* p' such that $\frac{1}{p} + \frac{1}{p'} = 1$.

2.2. **Hölder inequality.** Let $p \in [1, \infty]$, $f \in L^p$ and $g \in L^{p'}$. Then $fg \in L^1$ and

$$\int |fg| \leq \|f\|_{L^p} \|g\|_{L^{p'}}$$

2.3. **Minkowski's inequality.** Let $p \in [1, \infty]$ and $f, g \in L^p$. Then

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$$

2.4. **Interpolation inequality.** Let $1 \leq s \leq r \leq t \leq \infty$ and $f \in L^s \cap L^t$. Then $f \in L^r$ and

$$\|f\|_{L^r} \leq \|f\|_{L^s}^\theta \|f\|_{L^t}^{1-\theta}$$

where $\theta = \frac{s(t-r)}{r(t-s)}$.

2.6. **Theorem (Fischer-Riesz).** L^p is a complete normed vector space for $p \in [1, \infty]$.

2.7. **Riesz representation theorem.** Let $p \in [1, \infty)$ and let $\varphi : L^p \rightarrow \mathbf{R}$ be a linear continuous function. Then there exist a unique $u \in L^{p'}$ such that

$$\varphi(f) = \int u f \quad \forall f \in L^p$$

Thus, the topological dual of L^p is $L^{p'}$.

2.8. **Definition.** $L^p_{\text{loc}}(\Omega) :=$ functions such that $f\chi_K \in L^p(\Omega)$ for all compacts $K \subseteq \Omega$

2.9. **Fundamental lemma.** Let $f \in L^1_{\text{loc}}(\Omega)$ such that

$$\int f u = 0 \quad \forall u \in C_c(\Omega)$$

then $f = 0$ almost everywhere on Ω .

2.9. **Theorem (density).** The space $C_c(\Omega)$ is dense in $L^p(\Omega)$ for $1 \leq p < \infty$.

2.10. **Theorem (convolution in L^p).** The *convolution* of $f, g : \mathbf{R}^N \rightarrow \mathbf{R}$ is defined by

$$(f * g)(x) := \int_{\mathbf{R}} f(x-y)g(y)dy$$

It is well-defined in the following cases:

f	g	$f * g$
L^1	L^1	L^1
L^1	L^2	L^2
L^p	$L^{p'}$	$L^\infty \cap \mathcal{C}$

3. Hilbert spaces

3.1. **Definition.** A *real Hilbert space* is a real vector space H with an inner product (f, g) , such that H is complete under the norm $\|f\| := \sqrt{(f, f)}$. Examples: $\mathbf{R}^N$ and $L^2(\Omega)$.

3.2. **Cauchy-Schwarz inequality.** $|(f, g)| \leq \|f\| \|g\|$

3.3. **Pythagoras theorem.** If $(f, g) = 0$ then $\|f + g\|^2 = \|f\|^2 + \|g\|^2$.
If the sequence f_k is pairwise orthogonal then $\|\sum_k f_k\|^2 = \sum_k \|f_k\|^2$.

3.4. **Parallelogram identity.** $\|f\|^2 + \|g\|^2 = \frac{1}{2} (\|f + g\|^2 + \|f - g\|^2)$

3.5. **Polarization identity.** $(f, g) = \frac{1}{4} (\|f + g\|^2 - \|f - g\|^2)$

3.6. **Theorem (projection on a closed convex set).** Let $C \subseteq H$ be a nonempty closed convex set. Then, for any $f \in H$ there exist a unique $g \in C$ that minimizes the distance to f . The point g is characterized by the relation $\forall h \in C, (f - g, h - g) \leq 0$.

3.7. **Definition.** Let $A \subseteq H$. The *orthogonal* of A is the set

$$A^\perp := \{f, \forall a \in A, (f, a) = 0\}$$

It is a closed subspace of H .

3.8. **Theorem (orthogonal decomposition).** Let $F \subseteq H$ be a closed subspace. Then any element of H decomposes in a unique way in the form $f = g + h$ where $g \in F$ and $h \in F^\perp$. Moreover, g is the projection of f on F and h is the projection of f on $F^\perp$.

3.9. **Theorem (Riesz).** For any $f \in H$, the map $u \mapsto (u, f)$ is a continuous linear map from H to $\mathbf{R}$. Conversely, if $\varphi : H \rightarrow \mathbf{R}$ is a continuous linear map, there exists a unique $f \in H$ such that $\varphi(u) = (u, f) \quad \forall u \in H$.

Thus, the dual of H is isomorphic canonically to H .

3.10. **Definition.** A *Hilbert basis* of H is an orthonormal sequence of vectors $e_n, n \in \mathbf{N}$ which is total (it spans the whole space H).

3.11. **Theorem.** Any (separable) Hilbert space admits a Hilbert basis.

3.12. **Theorem ("Fourier" series).** Any element of f can be written in a unique way using a Hilbert basis

$$f = \sum_n c_n e_n.$$

The coefficients are $c_n = (f, e_n)$ and they satisfy **Parseval's identity** $\|f\|^2 = \sum_n \|c_n\|^2$.

3.13. **Corollary.** All (separable) Hilbert spaces are isomorphic (to $\ell^2(\mathbf{N})$).

3.14. **Continuity of linear forms.** Let $\lambda : H \rightarrow \mathbf{R}$ be a linear form. We say that is continuous if and only if there exists a constant C such that $|\lambda(f)| \leq C\|f\|$.

3.15. **Continuity of bilinear forms.** Let $a : H \times H \rightarrow \mathbf{R}$ be a bilinear form. We say that is continuous if and only if there exists a constant C such that $|a(f, g)| \leq C\|f\|\|g\|$. Moreover, if there exists a constant $c > 0$ such that $a(f, f) \geq c\|f\|^2$ then the form is said to be *coercive*.

3.16. **Theorem (Lax-Milgram, symmetric case).** Let E be an energy of the form

$$E(f) = \frac{1}{2} a(f, f) - b(f)$$

where a is a continuous bilinear, symmetric and coercive form, and b is a continuous linear form. Then there is a unique f that minimizes $E(f)$. Moreover, it is characterized by the relation

$$a(f, u) = b(u) \quad \forall u \in H.$$

4. Distributions

4.1. Definitions. Let $\Omega \subseteq \mathbf{R}$ be a connected subset. We introduce the following spaces

$$\mathcal{D}(\Omega) := \mathcal{C}_c^\infty(\Omega)$$

$$\mathcal{D}'(\Omega) := \text{dual of } \mathcal{D}(\Omega)$$

The dual is taken with respect to a topology of $\mathcal{D}$ defined elsewhere. The elements of $\mathcal{D}'$ are called *distributions*.

4.2. Notation. If $T : \mathcal{D} \rightarrow \mathbf{R}$ is a distribution, we use the following notations for $T(\varphi)$:

$$T(\varphi) = (T, \varphi) = \int_{\Omega} T \varphi = \int_{\Omega} T(x) \varphi(x) dx$$

4.3. Definition. We say that a sequence of distributions T_n converges to T if for any function $\varphi \in \mathcal{D}$ we have $(T_n, \varphi) \rightarrow (T, \varphi)$.

4.4. Examples.

- 1) To any function $f \in L^1_{\text{loc}}(\Omega)$ we associate a distribution T_f defined by

$$(T_f, \varphi) := \int_{\Omega} f(x) \varphi(x) dx.$$

This association is injective, and denoted simply by $T_f = f$.

- 2) Dirac delta: $(\delta, \varphi) := \varphi(0)$
 3) Derivative of Dirac delta: $(\delta', \varphi) := -\varphi'(0)$
 4) Dirac comb: $(\text{III}, \varphi) := \sum_{k \in \mathbf{Z}} \varphi(2\pi k)$.
 5) Principal value of $1/x$:

$$(\text{vp}(1/x), \varphi) := \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{\infty} \frac{\varphi(x) - \varphi(-x)}{x} dx$$

or equivalently, $\text{vp}(1/x) := \ln(|x|)'$ (defined below)

4.5. Derivative of a distribution. $(T', \varphi) := (T, -\varphi')$

4.6. Proposition. This definition of the derivative coincides with the classical one: when $T = T_f$ for a differentiable f , then $T'_f = T_{f'}$.

4.7. Product of a function and a distribution. $(fT, \varphi) := (T, f\varphi)$

Observation: the product of two distributions cannot be defined in general.

5. Sobolev spaces

5.1. Definition. We define the following spaces, for $1 \leq p \leq \infty$ and $k = 1, 2, 3, \dots$

$$W^{k,p}(\Omega) := \left\{ \begin{array}{l} \text{functions } f : \Omega \rightarrow \mathbf{R} \text{ such that} \\ \text{all their distributional derivatives} \\ \text{of order } 0, \dots, k \text{ belong to } L^p(\Omega) \end{array} \right\}$$

$$H^k(\Omega) := W^{k,2}(\Omega).$$

The space $W^{k,p}(\Omega)$ is endowed with the following norm (written here for the 1D case $\Omega \subseteq \mathbf{R}$):

$$\|f\|_{k,p} := \left(\|f\|_p^p + \|f'\|_p^p + \dots + \|f^{(k)}\|_p^p \right)^{1/p}$$

5.2. Properties. The spaces $W^{k,p}$ are complete normed vector spaces. The spaces H^k are Hilbert spaces (thus, their norms can be computed using a Hilbert basis and Parseval's identity).

6. Bibliography

- 6.1.** H. Brézis : *Analyse Fonctionnelle*
(canonical reference for functional analysis)
- 6.2.** L. C. Evans : *Partial Differential Equations*
(canonical reference for PDE)
- 6.3.** C. Gasquet, P. Witomski : *Analyse de Fourier et applications*
(nice exposition of the theory of distributions)

7. Exercices

- E1.** Find two counterexamples to Beppo Levi (missing the condition of monotonicity). One example using functions supported in the interval $[0,1]$ and another example using functions bounded by 1.
- E2.** Find two counterexamples to the dominated convergence theorem (missing the dominating function). One example using functions supported in the interval $[0,1]$ and another example using functions bounded by 1.
- *E3.** Can you find a function $f(x,y)$ that is integrable along each variable and $\int (\int f(x,y)dx) dy = 2$ but $\int (\int f(x,y)dy) dx = 3$.
- E4.** Find functions satisfying each of the following conditions (or prove that they do not exist)
- (a) $f \in L^{10}(\mathbf{R}), f \notin L^{12}(\mathbf{R})$
 - (b) $f \notin L^{10}(\mathbf{R}), f \in L^{12}(\mathbf{R})$
 - (c) $f \in L^{10}(\mathbf{R}), f \notin L^{12}(\mathbf{R}), f \in L^{14}(\mathbf{R})$
 - (d) $f \notin L^{10}(\mathbf{R}), f \in L^{12}(\mathbf{R}), f \notin L^{14}(\mathbf{R})$
- E5.** Prove that the dual of L^1 is L^∞ .
- **E6.** Prove that the dual of L^∞ is not L^1 .
- E7.** Find a Hilbert basis of $L^2([0,T])$ consisting of sine and cosine functions.
- E8.** Find a Hilbert basis of $L^2([0,T])$ consisting of only sine functions.
- E9.** Find a Hilbert basis of $L^2([0,T])$ consisting of only cosine functions.
- *E10.** Find a Hilbert basis of $L^2(\mathbf{R})$.
- E11.** The polarization identity allows to compute the scalar product from the associated norm. Does it imply that any normed space has an associated scalar product?
- E12.** Prove proposition 4.6 above.
- E13.** Define the shift of a distribution on $\mathbf{R}$ by a length a . (Analogous to the shift $f(x-a)$ of a function f).
- E14.** If $\tau_a T$ denotes the shift of T , prove that $T' = \lim_{h \rightarrow 0} \frac{T - \tau_h T}{h}$
- E15.** Write down the scalar product that leads to the Sobolev norm on section 6.
- E16.** Let $\Omega \subseteq \mathbf{R}^2$ and $f \in L^2(\Omega)$. Prove that there is a unique function u minimizing the energy

$$E(u) = \frac{1}{2} \int \|\nabla u(x,y)\|^2 dx dy - \int f(x,y)u(x,y) dx dy$$

and satisfying the condition $u = 0$ on $\partial\Omega$. Moreover, the optimal function satisfies Poisson equation $-\Delta u = f$. (Hint: define an appropriate Hilbert space, and apply Lax-Milgram theorem.)

8. Exercices (indications for the solution)

- E1.** Find two counterexamples to Beppo Levi (missing the condition of monotonicity). One example using functions supported in the interval $[0,1]$ and another example using functions bounded by 1.

The classical counterexamples are $f_n = n\mathbf{1}_{]0,1/n]}$ and $f_n = \mathbf{1}_{[n,n+1]}$. In both cases we have $\lim_n f_n(x) = 0$ and $\int f_n = 1$. None of these sequences of functions is monotonically increasing.

- E2.** Find two counterexamples to the dominated convergence theorem (missing the dominating function). One example using functions supported in the interval $[0,1]$ and another example using functions bounded by 1.

The same two examples of the previous exercise work here. None of these sequences of functions is dominated by an integrable function.

- *E3.** Can you find a function $f(x,y)$ that is integrable along each variable and $\int (\int f(x,y)dx) dy = 2$ but $\int (\int f(x,y)dy) dx = 3$.

The simplest solution may be the one defined by the following infinite table

3	-1	0	0	$\ddots$
0	1	-1	0	$\ddots$
0	0	1	-1	$\ddots$
0	0	0	1	$\ddots$
$\vdots$	$\ddots$	$\ddots$	$\ddots$	$\ddots$

The sum of each row is 0 except for the first one whose sum is 2. The sum of each column is 0 except for the first one whose sum is 3. Notice that you can interpret this as a function defined on $\mathbf{R}^+ \times \mathbf{R}^+$ which is piecewise constant in squares delimited by integer numbers.

The classical counterexample to Fubini's theorem with non-integrable functions is $f(x,y) = \frac{x^2-y^2}{(x^2+y^2)^2}$ defined for $(x,y) \in]0,1]^2$. Notice that $f(x,y) = \frac{\partial^2}{\partial x \partial y} \arctan(x,y)$, so that the integrals have values $\pm \frac{\pi}{4}$.

Notice that any such counterexample must necessarily have either vertical or horizontal asymptotes with positive and negative masses.

- E4.** Find functions satisfying each of the following conditions (or prove that they do not exist)

- (a) $f \in L^{10}(\mathbf{R}), f \notin L^{12}(\mathbf{R})$
- (b) $f \notin L^{10}(\mathbf{R}), f \in L^{12}(\mathbf{R})$
- (c) $f \in L^{10}(\mathbf{R}), f \notin L^{12}(\mathbf{R}), f \in L^{14}(\mathbf{R})$
- (d) $f \notin L^{10}(\mathbf{R}), f \in L^{12}(\mathbf{R}), f \notin L^{14}(\mathbf{R})$

First of all, case (c) is impossible due to the interpolation inequality given in the notes.

For the other cases, we can build counterexamples in a systematic way. Consider functions of the form

$$f_{\alpha,\beta}(x) = x^\alpha \mathbf{1}_{]0,1[}(x) + x^\beta \mathbf{1}_{[1,+\infty[}(x)$$

It is readily checked that $f_{\alpha,\beta} \in L^p$ if and only if $\alpha p + 1 > 0$ and $\beta p + 1 > 0$. By choosing appropriately the values of α and β we obtain functions belonging to the desired spaces.

- E5.** Prove that the dual of L^1 is L^∞ .

The inclusion $L^\infty \subseteq (L^1)'$ is a (trivial) particular case of Hölder's inequality. The other inclusion is a particular case of the Riesz representation theorem given above.

- **E6.** Prove that the dual of L^∞ is not L^1 .

The inclusion $L^1 \subseteq (L^\infty)'$ is still a particular case of Hölder's inequality. The other inclusion happens to be false. The actual construction of an element of $(L^\infty)'$ that does not correspond to integration against an integrable function is an advanced exercise:

To build such a linear form, we start with the space $\mathcal{C}_c$ of continuous, compactly supported functions on the real line. This is a vector subspace of L^∞ . The linear map $\varphi : f \mapsto f(0)$ is

defined on $\mathcal{C}_c$, it is continuous under the $\|\cdot\|_\infty$ norm and thus it can be extended linearly (in a non-unique way) to the whole of L^∞ . Let us call this extension $\bar{\varphi}$. By construction $\bar{\varphi} \in (L^\infty)'$.

Let us verify that there exists *no* function $u \in L^1$ such that $\bar{\varphi}(f) = \int u f \forall f \in L^\infty$. Assume, by contradiction, that such a function u exists. Then, for any function $f \in \mathcal{C}_c$ with $f(0) = 0$ we will have $\int u f = 0$, and by the fundamental lemma this implies $u = 0$ almost everywhere, so that $\bar{\varphi} = 0$.

- E7.** Find a Hilbert basis of $L^2([0, T])$ consisting of sine and cosine functions.

We start from the well-known fact that the family of functions $\{e^{in\theta}\}_{n \in \mathbb{Z}}$ is a—non-normalized—Hilbert basis of $L^2([0, 2\pi])$ (this is the main result of the theory of Fourier series, and it follows from Fejér theorem and the density of continuous functions inside L^2).

Now, write the whole space as an orthogonal sum

$$L^2([0, 2\pi]) = H_0 \oplus H_1 \oplus H_2 \oplus \cdots$$

where H_0 is the space of constant functions and H_n is the two-dimensional space spanned by $e^{\pm in\theta}$. Due to Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, the space H_n is also spanned by the two functions $\cos(n\theta)$ and $\sin(n\theta)$. By linearly rescaling the argument $\theta = \frac{2\pi t}{T}$ we obtain a (non-normalized) basis for $L^2([0, T])$ consisting of sine and cosine functions:

$$\left\{ \cos \frac{2\pi n t}{T} \right\}_{n=0,1,2,\dots} \cup \left\{ \sin \frac{2\pi n t}{T} \right\}_{n=1,2,3,\dots}$$

This is the classical Fourier basis of $L^2([0, T])$.

- E8.** Find a Hilbert basis of $L^2([0, T])$ consisting of only sine functions.

By the argument above, the set of functions

$$\left\{ \cos \frac{\pi n t}{T} \right\}_{n=0,1,2,\dots} \cup \left\{ \sin \frac{\pi n t}{T} \right\}_{n=1,2,3,\dots}$$

is a basis of $L^2([-T, T])$. Since the domain is symmetric around zero, this space can be written as the orthogonal sum of even and odd functions

$$L^2([-T, T]) = E \oplus O$$

so that $\left\{ \cos \frac{\pi n t}{T} \right\}_{n=0,1,2,\dots}$ is a basis of E and $\left\{ \sin \frac{\pi n t}{T} \right\}_{n=1,2,3,\dots}$ is a basis of O .

Now, since any function defined on $[0, T]$ can be extended to an odd function defined on $[-T, T]$, we see that the set of functions

$$\left\{ \sin \frac{\pi n t}{T} \right\}_{n=1,2,3,\dots}$$

must be a Hilbert basis of $L^2([0, T])$.

Notice that all the functions on this set vanish at the boundaries of the interval. Nevertheless, they are still able to approximate functions without this property, such as constants (by approximating the oddly-extended constant on $[-T, T]$). This basis is called the *Dirichlet basis* of $L^2([0, T])$.

- E9.** Find a Hilbert basis of $L^2([0, T])$ consisting of only cosine functions.

By an analogous argument as on the previous exercise we see that the set of functions

$$\left\{ \cos \frac{\pi n t}{T} \right\}_{n=0,1,2,\dots}$$

is a Hilbert basis of $L^2([0, T])$.

Notice that all the functions on this set have vanishing derivative at the boundaries of the interval. Nevertheless, they are still able to approximate functions without this property, such as $f(t) = t$ (by approximating the symmetrically-extended function on $[-T, T]$). This basis is called the *Neumann basis* of $L^2([0, T])$.

- *E10.** Find a Hilbert basis of $L^2(\mathbb{R})$.

There are many possible approaches:

- (1) The ugly but direct and easy way: take a Hilbert basis $\{e_n\}_{n \in \mathbf{Z}}$ of $L^2([0,1])$ and build the family of functions

$$e_{m,n}(x) = e_n(x - m)\mathbf{1}_{[m,m+1]}(x) \quad m, n \in \mathbf{Z}$$

- (2) Haar wavelets: consider the “mother” wavelet $\varphi(x) = \begin{cases} 1 & x \in [0, \frac{1}{2}[\\ -1 & x \in [\frac{1}{2}, 1[\\ 0 & \text{otherwise} \end{cases}$

and build the family of functions

$$\varphi_{m,n}(x) = 2^m \varphi(2^m x - n) \quad m, n \in \mathbf{Z}$$

It is readily checked that this family is orthogonal. It spans $L^2(\mathbf{R})$ because piecewise-constant functions of compact support are dense in $L^2(\mathbf{R})$ (even if we take “pieces” whose ends occur at integer powers of two).

- (3) Hermite functions: The Fourier transform is a self-adjoint, positive, map $\widehat{\cdot}: L^2(\mathbf{R}) \rightarrow L^2(\mathbf{R})$. Its eigenfunctions are the Hermite functions

$$\varphi_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2}$$

that form a Hilbert basis of $L^2(\mathbf{R})$.

- E11.** The polarization identity allows to compute the scalar product from the associated norm. Does it imply that any normed space has an associated scalar product?

No. If we try to define

$$\langle f, g \rangle := \frac{1}{4} (\|f + g\|^2 - \|f - g\|^2)$$

there’s no reason for this expression to be linear on the first argument, for example. An “inner product” thus defined will be an actual inner product only when the original norm satisfies the parallelogram identity.

- E12.** Prove proposition 4.6 above.

This is a direct check of the definitions. Let f be a locally integrable function. We want to check that for any $\varphi \in \mathcal{D}$ we have $\langle (T_f)', \varphi \rangle = \langle T_{f'}, \varphi \rangle$. Now, developing the first term:

$$\begin{aligned} \langle (T_f)', \varphi \rangle &= \langle T_f, -\varphi' \rangle && \text{by definition of derivative of a distribution} \\ &= - \int_{-\infty}^{+\infty} f(x) \varphi'(x) dx && \text{by definition of distribution associated to a function} \\ &= - \int_{-M}^{+M} f(x) \varphi'(x) dx && \text{because } \varphi \text{ is compactly supported} \\ &= [f(x) \varphi(x)]_{-M}^M + \int_{-M}^M f'(x) \varphi(x) dx && \text{integration by parts} \\ &= \int_{-\infty}^{\infty} f'(x) \varphi(x) dx && \text{because } \varphi \text{ is compactly supported} \\ &= \langle T_{f'}, \varphi \rangle && \text{by definition of distribution associated to a function} \end{aligned}$$

- E13.** Define the shift of a distribution on $\mathbf{R}$ by a length a . (Analogous to the shift $f(x - a)$ of a function f).

The shift of a function is defined as $(\tau_a f)(x) := f(x - a)$. For any distribution $T \in \mathcal{D}'$ and any $a \in \mathbf{R}$ we define a distribution $\tau_a T$ by

$$\langle \tau_a T, \varphi \rangle := \langle T, \tau_{-a} \varphi \rangle$$

Notice that this definition makes sense because $\tau_{-a} \varphi \in \mathcal{D}$. It is also natural because it extends the shift of functions: $\tau_a T_f = T_{(\tau_a f)}$, as can be readily checked.

E14. If $\tau_a T$ denotes the shift of T , prove that $T' = \lim_{h \rightarrow 0} \frac{T - \tau_h T}{h}$

Pour vérifier que deux distributions sont égales, il suffit de voir que son action sur toute fonction $\varphi \in \mathcal{D}$ donne le même résultat. D'un côté on a

$$\left\langle \lim_{h \rightarrow 0} \frac{\tau_{-h} T - T}{h}, \varphi \right\rangle = \lim_{h \rightarrow 0} \left\langle \frac{\tau_{-h} T - T}{h}, \varphi \right\rangle$$

par définition de limite de distributions,

$$= \lim_{h \rightarrow 0} \frac{\langle \tau_{-h} T, \varphi \rangle - \langle T, \varphi \rangle}{h}$$

par linéarité,

$$= \lim_{h \rightarrow 0} \frac{\langle T, \tau_h \varphi \rangle - \langle T, \varphi \rangle}{h}$$

par définition de τ ,

$$= \lim_{h \rightarrow 0} \left\langle T, \frac{\tau_h \varphi - \varphi}{h} \right\rangle$$

par linéarité,

$$= \left\langle T, \lim_{h \rightarrow 0} \frac{\tau_h \varphi - \varphi}{h} \right\rangle$$

par continuité de T ,

$$= \langle T, -\varphi' \rangle$$

par définition de dérivée de $\varphi \in \mathcal{C}^\infty$,

$$= \langle T', \varphi \rangle$$

par définition de dérivée d'une distribution.

E15. Write down the scalar product that leads to the Sobolev norm on section 6.

$$\langle f, g \rangle_{H^k} = \langle f, g \rangle_{L^2} + \langle f', g' \rangle_{L^2} + \cdots + \langle f^{(k)}, g^{(k)} \rangle_{L^2}$$

E16. Let $\Omega \subseteq \mathbf{R}^2$ and $f \in L^2(\Omega)$. Prove that there is a unique function u minimizing the energy

$$E(u) = \frac{1}{2} \int \|\nabla u(x, y)\|^2 dx dy - \int f(x, y) u(x, y) dx dy$$

and satisfying the condition $u = 0$ on $\partial\Omega$. Moreover, the optimal function satisfies Poisson equation $-\Delta u = f$. (*Hint: define an appropriate Hilbert space, and apply Lax-Milgram theorem.*)

(*sketch of the solution*) The exercice is missing the hypothesis that Ω is bounded. Consider the Sobolev space $H_0^1(\Omega)$ of weakly derivable functions whose gradient belongs to $L^2(\Omega)$ and that vanish on the boundary $\partial\Omega$. The bilinear form $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$ is coercive (due to Poincaré inequality on the domain Ω) and continuous. The linear form $b(u) = \int_{\Omega} f u$ is continuous. Thus we can apply Lax-Milgram theorem to obtain existence and uniqueness of a minimizer for E . The condition satisfied by the minimum u is $a(u, v) = b(v) \forall v \in H_0^1(\Omega)$. Integrating by parts and applying the fundamental lemma this condition can be rewritten as Poisson equation $-\Delta u = f$.

Note that Poisson equation can be obtained formally from the energy by identifying the lagrangian $L = \frac{1}{2} (u_x^2 + u_y^2) - f u$, thus the first variation is $\frac{\delta E}{\delta u} = \frac{\partial L}{\partial u} - \frac{d}{dx} \frac{\partial L}{\partial u_x} - \frac{d}{dy} \frac{\partial L}{\partial u_y}$ and the Euler-Lagrange equations are $\frac{\delta E}{\delta u} = 0$ which is Poisson equation.